Topological order and Long range entanglements

vertex algebra and tensor category in condensed matter physics

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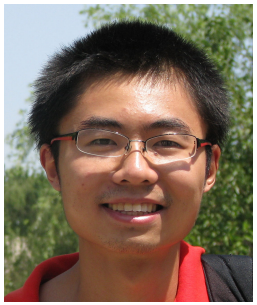
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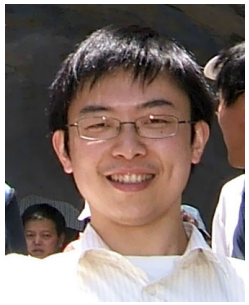
Z.-C. Gu

Physical motivations

- A quantum system is LOCAL hermitian operator acting quantum states which are complex wave functions of N variables $\Psi(z_1, ..., z_N)$: $\mathbb{C}^N \to \mathbb{C}$.
- A quantum phase is a equivalent class of the wave functions $\{\Psi_1(z_1, ..., z_N), \Psi_2(z_1, ..., z_N), ... | \Psi_1 \sim \Psi_2\}$

- Understand quantum phases
  is to understand all those equivalent classes of the wave functions
  and
  is to classify wave functions of infinite variables.

How to classify symm. polynomials of infinite variables?

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We will use fraction quantum Hall (FQH) states to motivate our definition.

- Define what are *symmetric polynomials of infinite variables*
- Find an equivalence relation $P_1 \sim P_2$
- The equivalent class of the equivalence relation classify the symmetric polynomials of infinite variables.

FQH states = symm. polynomials with inf. variables

- 1-electron state on a sphere with N_ϕ magnetic flux quanta:

$$P(z) = \sum_{m=0}^{N_\phi} c_m z^m, \ (\tilde{P}, P) = \int \frac{d^2 z}{1+|z|^2}^{2+N_\phi} \tilde{P}^*(z) P(z) \ \to \mathcal{H}_1$$

- N-electron state on a sphere with N_ϕ magnetic flux quanta

$$P(z_1, ..., z_N) = \sum_{m_1,..,m_N} c_{m_1,..,m_N} z_1^{m_1} ... z_N^{m_N} \ \in \mathcal{H}_N \equiv \mathcal{H}_1^N$$

  where $P(z_1, ..., z_N)$ is symmetric.
- Hamiltonian $\hat{H} = \sum_{i<j} V_g(z_i - z_j)$ and its matrix $H_{ij} = (P_i, \hat{H} P_j)$.
- There is one minimal energy eigenfunction P_0 whose form is determined by V_g. The energy gap to the first excited state is Δ.
- $N \to \infty$ limit: If there exists a sequence of electron numbers $N_1, N_2, ..., \infty$ and the corresponding $N_{\phi,i} = \nu^{-1} N_i + p$ such that the corresponding gap $\lim_{i\to\infty} \Delta_i = \Delta_\infty \neq 0$ then the corresponding sequence P_i is a *symmetric polynomial of infinite variables*.

The equivalence relation $\rightarrow$ phases of FQH states

- Consider a smooth path from V_g to $V_{g'}$. It induces a path from P_g to $P_{g'}$. If $\Delta_\infty \neq 0$ along the path, then $P_g \sim P_{g'}$
  $\rightarrow$ A classification of symmetric polynomials of infinite variables
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- How to label all the different classes
- Given the label of a class, how to calculate the universal properties, such as non-Abelian statistics, fraction charge, ...

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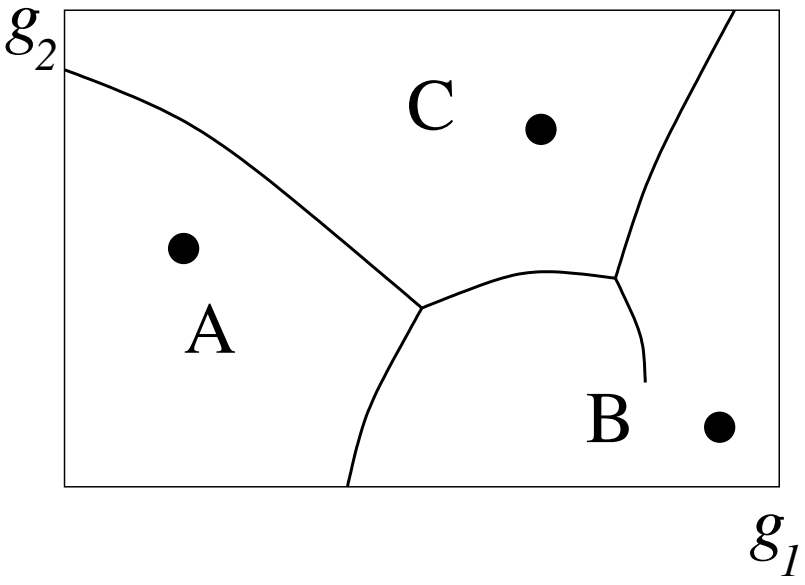
A physical definition:

- Hamiltonian H on a genus g Riemann surface $\to$ degenerate ground states: degeneracy $= N_{deg}(g)$.
- As we change the shape of the Riemann surface $\to$ fiber bundle on the moduli space.
- Flat connection $\to$ projective representations of the modular transformation S, T.

$(N_{deg}(g), S, T)$ *describe* **topological order** *in FQH phases.* Wen 89

How to find labels for different classes?

Study "ideal" wave functions (or ideal polynomials).
Find the labels through ideal polynomials.



Three FQH states: the exact zero-energy ground states

- $\nu = 1/2$ Laughlin state: $z_1 \approx z_2$, second-order zero

$$P_{1/2} = \prod_{i<j} (z_i - z_j)^2, \quad V_{1/2}(z_1, z_2) = \delta(z_1 - z_2),$$

$$[\sum_{i<j} V_{1/2}(z_i - z_j)] P_{1/2} = 0.$$

All other states have finite energies in $N \to \infty$ limit (gapped).

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$$P_{1/4} = \prod_{i<j} (z_i - z_j)^4$$

$$V_{1/4}(z_1, z_2) = v_0 \delta(z_1 - z_2) + v_2 \partial^2_{z_1^*} \delta(z_1 - z_2) \partial^2_{z_1}$$

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- $\nu = 1$ Pfaffian state: $z_1 \approx z_2$, no zero; $z_1 \approx z_2 \approx z_3$, second-order zero;

$$P_{Pf} = \mathcal{A}\Big(\frac{1}{z_1 - z_2} \frac{1}{z_3 - z_4} \cdots \frac{1}{z_{N-1} - z_N}\Big) \prod_{i<j} (z_i - z_j) = \mathrm{Pf}(\frac{1}{z_i - z_j}) \prod_{i<j} (z_i - z_j)$$

$$V_{Pf}(z_1, z_2, z_3) = \mathcal{S}[v_0 \delta(z_1 - z_2) \delta(z_2 - z_3) - v_1 \delta(z_1 - z_2) \partial_{z_3^*} \delta(z_2 - z_3) \partial_{z_3}]$$

Summary

Different interaction $V(z_i - z_j)$ in

$$H = \sum_{i=1}^{N} -(\partial_z - \frac{B}{4} z^*)(\partial_{z^*} + \frac{B}{4} z) + \sum_{i<j} V_(z_i - z_j)$$

$\rightarrow$ different FQH states:

- $z_1 \approx z_2$, second-order zero $P_{1/2} = \prod_{i<j}(z_i - z_j)^2$.
- $z_1 \approx z_2$, fourth-order zero $P_{1/4} = \prod_{i<j}(z_i - z_j)^4$.
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How to understand other FQH states?
How to systematically label them?

Pattern of zeros

Let $z_1, ..., z_a \to z^{(a)}$ through $z_i = \lambda \xi_i + z^{(a)}$, $i = 1, 2, \cdots, a$

$$P(\{z_i\}) = \lambda^{S_a} P(\xi^1, ..., \xi^a; z^{(a)}, z_{a+1}, z_{a+2}, \cdots) + O(\lambda^{S_a+1})$$

- The sequence of integers $\{S_a\}$ characterizes the polynomial $P(\{z_i\})$ and is called the pattern of zeros.
- $\nu = 1/2$ Laughlin state $S_1, S_2, \cdots$: $0, 2, 6, 12, 20, 30, 42, 56, \cdots$.

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- $\nu = 1/2$ Laughlin state $S_1, S_2, \cdots$: $0, 2, 6, 12, 20, 30, 42, 56, \cdots$.
- **Unique fusion cond.**: P does not depend on the "shape" $\{\xi^i\}$

$$P(\{\xi^i\}; z^{(a)}, z_{a+1}, z_{a+2}, \cdots) \propto P(z^{(a)}, z_{a+1}, z_{a+2}, \cdots)$$

- **Pattern of zeros and orbital/occupation distribution**

  S_N is the total power of z_i if the polynomial has N variables.

  Relation to monomial symmetric polynomial ("highest weight"?):

  Let $l_a = S_a - S_{a-1}$ or $S_a = \sum_{i=1}^a l_i$, then

$$P(\{z_i\}) \sim \mathcal{S}[z_1^{l_1} z_2^{l_2} \cdots] + \cdots, \quad = m_{\{l_1, l_2, ...\}}(z_i, .., z_N) + ...$$

The pattern of zero of $\nu = 1/2$ Laughlin state is also described by

$$l_1, l_2, \cdots : 0, 2, 4, 6, 8, 10, \cdots$$

$$n_0 n_1 n_2 \cdots : 1010101010101010 \cdots$$

For P_1:

$$\prod_{i<j}(z_i - z_j) = \mathcal{A}[(z_1)^0(z_2)^1 ...] \to S_1, S_2, ... = 0, 1, 3, 6, 10, ...$$

$$l_1, l_2, \cdots : 0, 1, 2, 3, 4, \cdots , \qquad n_0 n_1 n_2 \cdots : \ 111111111111 \cdots$$

For $P_{1/2}$:

$$\prod_{i<j}(z_i - z_j)^2 = \Big(\mathcal{A}[(z_1)^0(z_2)^1(z_3)^2 ...]\Big)^2 \to S_1, S_2, ... = 0, 2, 6, 12, 20, ...$$

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root monomial poly. $= \mathcal{S}[(z_1)^0(z_2)^2(z_3)^4 ...]$

- $\nu = 1/4$ Laughlin state $P_{1/4} = \prod_{i<j}(z_i - z_j)^4$

$$S_1, S_2, \cdots : 0, 4, 12, 24, 40, 60, 84, \cdots$$

$$l_1, l_2, \cdots : 0, 4, 8, 12, 16, 20, \cdots$$

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$$\text{root monomial poly.} \quad = \mathcal{S}[(z_1)^0 (z_2)^4 (z_3)^8 ...]$$

A cluster (unit cell): 1 particles 4 orbitals

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$$S_1, S_2, \cdots : 0, 0, 2, 4, 8, 12, 18, 24, \cdots$$

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$$n_0 n_1 n_2 \cdots : 2020202020202020202 \cdots$$

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- FQH $\leftrightarrow$ 1D pattern (by considering FQH state on thin cylinder)

Haldane & Rezayi, 94; Seidel & Lee, 06; Bergholtz, Kailasvuori, Wikberg, Hansson, Karlhede, 06; Bernevig & Haldane, 07

A classification problem

- We have seen that
  each symmetric polynomial $P(\{z_i\}) \to \{S_a\}$ a pattern of zeros.
  But each sequence of integers $\{S_a\} \not\to P(\{z_i\})$
- Find all the conditions a sequence $\{S_a\}$ must satisfy, such that $\{S_a\}$ describe a symmetric polynomial that satisfies the unique fusion condition. $\to$
  A classification of symmetric polynomials (FQH states) through pattern of zeros.

Conditions on pattern of zeros – ground state

- Concave conditions

$$S_{a+b} - S_a - S_b \geq 0,$$

$$S_{a+b+c} - S_{a+b} - S_{b+c} - S_{a+c} + S_a + S_b + S_c = \text{even} \geq 0$$

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- n-cluster condition (quasi-periodic in S_a and periodic in n_l):

$$S_{a+kn} = S_a + kS_n + \frac{k(k-1)nm}{2} + kma$$

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- $(m; S_2, \cdots, S_n)$ that satisfy the above conditions correspond to *good* symmetric polynomials. $\rightarrow$ Those $(m; S_2, \cdots, S_n)$ "classify" symmetric polynomials and FQH states (with $\nu = n/m$).

Primitive solutions for pattern of zeros

The conditions are semi-linear $\rightarrow$
if $(m; S_2, S_3, \cdots)$ and $(m'; S'_2, S'_3, \cdots)$ are solutions, then $(m''; S''_2, S''_3, \cdots) = (m; S_2, S_3, \cdots) + (m'; S'_2, S'_3, \cdots)$ is also a solution $\sim P''(\{z_i\}) = P(\{z_i\})P'(\{z_i\})$

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1-cluster state: $\nu = 1/m$ Laughlin state

$$P_{1/m}: \qquad \mathbf{S} = (m;),$$
$$(n_0, \cdots, n_{m-1}) = (1, 0, \cdots, 0).$$

2-cluster state: Pfaffian state (Z_2 parafermion state)

$$P_{\frac{2}{2};Z_2}: \qquad (m; S_2) = (2; 0),$$
$$(n_0, \cdots, n_{m-1}) = (2, 0)$$

3-cluster state: Z_3 parafermion state

$$P_{\frac{3}{2};Z_3}: \qquad (m; S_2, S_3) = (2; 0, 0),$$
$$(n_0, \cdots, n_{m-1}) = (3, 0)$$

4-cluster state: Z_4 parafermion state

$$P_{\frac{4}{2};Z_4} : (m; S_2, \cdots, S_n) = (2; 0, 0, 0),$$
$$(n_0, \cdots, n_{m-1}) = (4, 0),$$

5-cluster states: Z_5 (generalized) parafermion state

$$P_{\frac{5}{2};Z_5} : (m; S_2, \cdots, S_n) = (2; 0, 0, 0, 0),$$
$$(n_0, \cdots, n_{m-1}) = (5, 0)$$

$$P_{\frac{5}{8};Z_5^{(2)}} : (m; S_2, \cdots, S_n) = (8; 0, 2, 6, 10),$$
$$(n_0, \cdots, n_{m-1}) = (2, 0, 1, 0, 2, 0, 0, 0)$$

6-cluster state:

$$P_{\frac{6}{2};Z_6} : (m; S_2, \cdots, S_n) = (2; 0, 0, 0, 0, 0),$$
$$(n_0, \cdots, n_{m-1}) = (6, 0)$$

7-cluster states:

$$P_{\frac{7}{2};Z_7} : (m; S_2, \cdots, S_n) = (2; 0, 0, 0, 0, 0, 0),$$
$$(n_0, \cdots, n_{m-1}) = (7, 0)$$

$$P_{\frac{7}{8};Z_7^{(2)}} : (m; S_2, \cdots, S_n) = (8; 0, 0, 2, 6, 10, 14),$$
$$(n_0, \cdots, n_{m-1}) = (3, 0, 1, 0, 3, 0, 0, 0)$$

$$P_{\frac{7}{18};Z_7^{(3)}} : (m; S_2, \cdots, S_n) = (18; 0, 4, 10, 18, 30, 42),$$
$$(n_0, \cdots, n_{m-1}) = (2, 0, 0, 0, 0, 1, 0, 0, 0, 2, 0, 0, 0, 0, 0)$$

$$P_{\frac{7}{14};C_7} : (m; S_2, \cdots, S_n) = (14; 0, 2, 6, 12, 20, 28),$$
$$(n_0, \cdots, n_{m-1}) = (2, 0, 1, 0, 1, 0, 1, 0, 2, 0, 0, 0, 0, 0)$$

- Also get composite parafermion state $P = P_{Z_{n_1}} P_{Z_{n_2}}$

How good is the pattern-of-zero classification

Not so good :-(

- Every symm. poly. P corresponds to a unique pattern of zero $\{S_a\}$. Only some patterns of zero correspond to a unique symm. poly.
- It appears that all the primitive patterns of zero correspond to a unique symm. poly.
- It is known that
  some composite patterns of zero $\rightarrow$ a unique symm. polynomial
  some composite patterns of zero $\rightarrow$ several symm. polynomials
  So in general, we need more information than $\{n; m; S_a\}$ to fully characterize symmetry polynomial of infinite variables

Topological properties from pattern of zeros

For those patterns of zeros that uniquely characterize a FQH wave function, we should be able to calculate the topological properties of FQH states from the data $(n; m; S_2, \cdots, S_n)$.

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For those patterns of zeros that uniquely characterize a FQH wave function, we should be able to calculate the topological properties of FQH states from the data $(n; m; S_2, \cdots, S_n)$.

- **Filling fraction** $\nu = \frac{N_\infty}{N_{\phi,\infty}} = n/m$.
- **Topological degeneracy on torus (and other Riemann surface)**
- **Number of quasiparticle types (objects in tensor category)**
- **Quasiparticle charges**
- Quasiparticle scaling dimensions
- **Quasiparticle fusion algebra (fusion of objects)**
- Quasiparticle statistics (Abelian and non-Abelian)
- The counting of edge excitations (central charge c and spectrum)

Topological properties = modular tensor category + ...

- FQH states are classified by modular tensor category + ...
- Symmetric polynomials of infinite variables are classified by modular tensor category + ...

Topological degeneracy

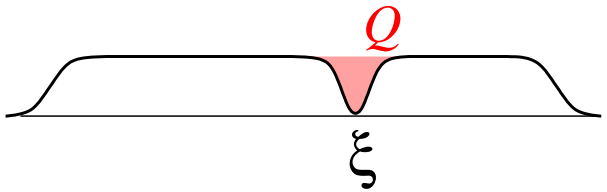
- Symmetric polynomial $\rightarrow$ FQH state on plane/sphere.

  Holomorphic function with quasi-periodic boundary condition in x- and y-directions (like the θ-functions) $\rightarrow$ FQH state on torus.
- How many holomorphic functions with a pattern of zero $(n; m; S_a)$ can be put on a torus?
  How many zero-energy states the ideal FQH Hamiltonian have on a torus?

We should be able to calculate the topological degeneracy from the data $(n; m; S_a)$.

A quasiparticle

- A quasiparticle is a defect in the ground state wave function $P(\{z_i\})$. It is a place where we have more power of zeros.
- For example:
  Ground state $P_{1/2} = \prod_{i<j}(z_i - z_j)^2$.
  A quasiparticle state $P_\xi = \prod_i (z_i - \xi) \prod_{i<j}(z_i - z_j)^2$
- Density distribution of a quasiparticle state:

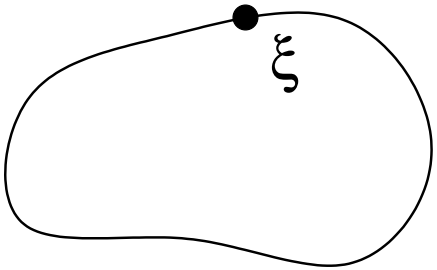


The defect at η is point-like and carries a (fractional) charge Q.

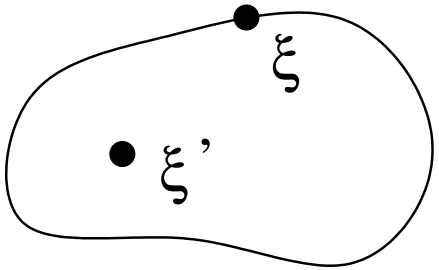
Fractional statistics = Berry phase of exch. quasi-holes

The quasiparticle wave function P_ξ defines a section of holomorphic line bundle on the surface $\Sigma = \{\xi\}$.

The Berry phase $= \oint d\xi \cdot \mathbf{a}(\xi)$



Berry's phase =2 π x encl. elec.



Berry's phase =2 π x (encl. elec. – 1/m)

- Statistical angle $\theta = -\pi/m$

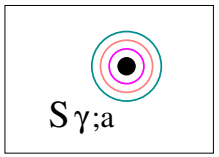
Quasiparticle = object in tensor category system.

Quasiparticles form a modular tensor category (MTC)

How to calculate MTC from the pattern of zeros?

The pattern of zeros for quasiparticles

- A quasiparticle γ in a FQH state can also be quantitatively characterized by pattern of zeros $\{S_{\gamma;a}\}$:



  Let $\Psi_\gamma(\xi, z_i)$ be a FQH wave function with a quasiparticle γ at ξ, then $S_{\gamma;a}$ is the order of zero of $\Psi_\gamma(\xi, z_i)$ when we bring a electrons to ξ. ($\Psi_\gamma(\xi, z_i) = \prod(z_i - \xi)\Psi_{1/m}$ for a Laughlin quasiparticle.)
- $\{S_{\gamma;a}\}$ must satisfies:

$$S_{\gamma;a+b} - S_{\gamma;a} - S_b \geq 0,$$
$$S_{\gamma;a+b+c} - S_{\gamma;a+b} - S_{\gamma;a+c} - S_{b+c} + S_{\gamma;a} + S_b + S_c \geq 0$$

- The above equations have many solutions, and each solution correspond to a type of quasiparticle.

  $\{S_{\gamma;a}\}$ **are quantitative characterizations of the quasiparticles. Such quantitative characterizations allow us to calculation quasiparticle properties quantitatively.**

The pattern of zeros for quasiparticles

For the $\nu = 1$ Pfaffian state ($n = 2$ and $m = 2$)

$$S_1, S_2, \cdots : 0, 0, 2, 4, 8, 12, 18, 24, \cdots$$

$$n_0 n_1 n_2 \cdots : 2020202020202020202 \cdots$$

- Quasiparticle solutions ($S_{\gamma;a} \to l_{\gamma;a} \to n_{\gamma;l}$):

$$n_{\gamma;0} n_{\gamma;1} n_{\gamma;2} \cdots : 2020202020202020202 \cdots \qquad Q_\gamma = 0$$

$$n_{\gamma;0} n_{\gamma;1} n_{\gamma;2} \cdots : 0202020202020202020 \cdots \qquad Q_\gamma = 1$$

$$n_{\gamma;0} n_{\gamma;1} n_{\gamma;2} \cdots : 1111111111111111111 \cdots \qquad Q_\gamma = 1/2$$

Unit cell: m orbitals + n electrons

- All other quasiparticle solutions can obtained from the above three by removing extra electrons $\to$ only 3 quasiparticle types.

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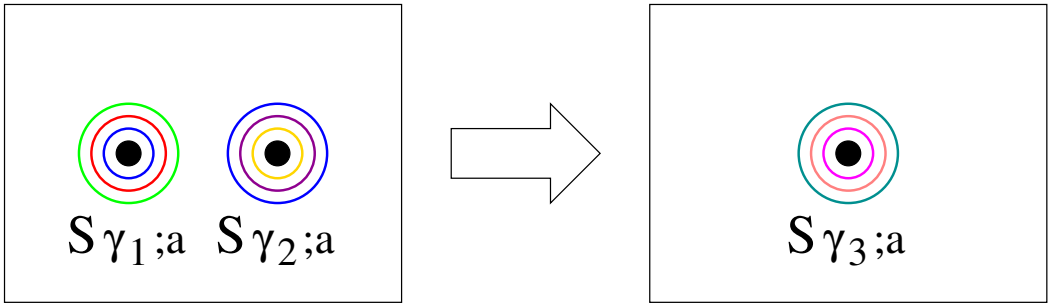
- *Ground state degeneracy on torus = number of quasiparticle types*
- *Charge of quasiparticles*

$$Q_\gamma = \frac{1}{m} \sum_{a=1}^{n} (l_{\gamma;a} - l_a) = \frac{S_{\gamma;n} - S_n}{S_{n+1} - S_n}$$

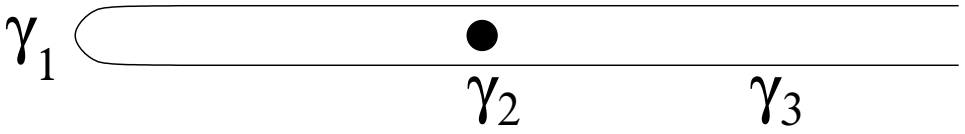
Quasiparticle fusion algebra: $\gamma_1\gamma_2 = \sum_{\gamma_3} N^{\gamma_3}_{\gamma_1\gamma_2}\gamma_3$

Consider a particular fusion channel $\gamma_1\gamma_2 \to \gamma_3$



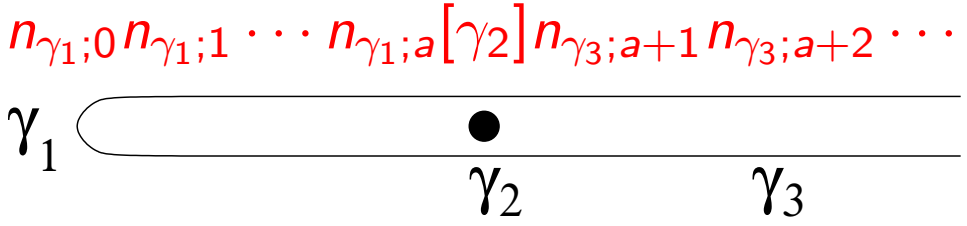
Its occupation representation is a "domain wall" Ardonne etc, 2008

$$n_{\gamma_1;0}\, n_{\gamma_1;1} \cdots n_{\gamma_1;a}[\gamma_2]\, n_{\gamma_3;a+1}\, n_{\gamma_3;a+2} \cdots$$



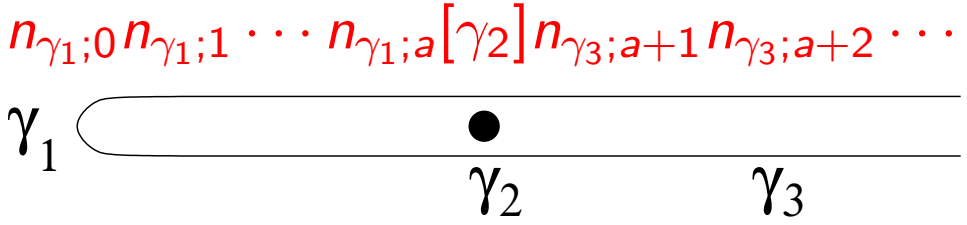
From the domain wall, we can see $n_{\gamma_1;l}$ and $n_{\gamma_3;l}$, but we do not know $n_{\gamma_2;l}$.

Quasiparticle fusion algebra: $\gamma_1\gamma_2 = \sum_{\gamma_3} N^{\gamma_3}_{\gamma_1\gamma_2}\gamma_3$



- The key is to find a condition on $n_{\gamma_2;l}$ so that it can induce a domain wall between $n_{\gamma_1;l}$ and $n_{\gamma_3;l}$

Quasiparticle fusion algebra: $\gamma_1\gamma_2 = \sum_{\gamma_3} N^{\gamma_3}_{\gamma_1\gamma_2}\gamma_3$



- The key is to find a condition on $n_{\gamma_2;l}$ so that it can induce a domain wall between $n_{\gamma_1;l}$ and $n_{\gamma_3;l}$: Barkeshli & Wen, 2008

$$\sum_{j=1}^{b} l^{sc}_{\gamma_2+c;j} \le \sum_{j=1}^{b} \left(l^{sc}_{\gamma_3+a+c;j} - l^{sc}_{\gamma_1+a;j} + l^{sc}_{j} \right)$$

for any $a, b, c \in Z_+$, where $l^{sc}_{\gamma;a} = l_{\gamma;a} - \frac{m(Q_\gamma+a-1)}{n}$

- *Solving the above equation* $\to N^{\gamma_3}_{\gamma_1\gamma_2}$
- *Ground state degeneracies on any Riemann surfaces.*

Relation to Vertex algebra

$$P(\{z_i\}) = \langle \prod_i V_e(z_i) \rangle, \quad V_a(z) \quad = (V_e)^a, \quad V_a V_b = V_{a+b}.$$

The electron cluster operator V_a form a Z-graded vertex algebra:

$$V_a(z) V_b(w) = \frac{C_{ab}}{(z-w)^{h_a+h_b-h_{a+b}}} V_{a+b}(w) + \ldots$$

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- A relation between the h_a and S_a (h_a is the dimension of V_a)

$$h_{a+b} - h_a - h_b = S_{a+b} - S_a - S_b$$

n-cluster condition implies that $h_a \propto a^2$ if $a = 0 \bmod n$.

$$h_a = S_a - \frac{aS_n}{n} + \frac{am}{2}$$

More complete characterization of FQH wave functions (symmetric polynomials) is given by $(n; m; S_a; C_{ab}, \ldots)$**.**

Z_n-vertex algebra

The Z-graded vertex algebra $=$ Z_n-vertex algebra $\otimes$ $U(1)$ current algebra:

$$V_a(z) = \psi_a(z) e^{ia\phi(z)\sqrt{m/n}}$$

where $j = \partial\phi$ generates the $U(1)$ current algebra and ψ_a generates the Z_n-vertex algebra:

$$\psi_a(z)\psi_b(w) = \frac{C_{ab}}{(z-w)^{h_a^{sc}+h_b^{sc}-h_{a+b}^{sc}}}\psi_{a+b}(w) + \ldots$$

$$\psi_n = 1, \text{ the n-cluster condition.}$$

where the scaling dimension of $\psi_a(z)$ is

$$h_a^{sc} = h_a - \frac{a^2 m}{2n} = S_a - \frac{aS_n}{n} + \frac{am}{2} - \frac{a^2 m}{2n}, \qquad h_a^{sc} = h_{a+n}^{sc}$$

Does the data $(n; m; h_a^{sc}, C_{ab}, \ldots)$ completely characterize the Z_n-vertex algebra and symmetric polynomial?

Classify Z_n-vertex algebra= classify symmetric polynomials

- From the pattern-of-zero consideration, we get concave conditions

$$S_a = h_a^{sc} - a h_1^{sc} + \frac{a(a-1)m}{2n} = \text{integer} \geq 0$$

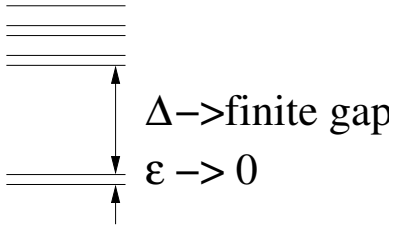
$$h_{a+b}^{sc} - h_a^{sc} - h_b^{sc} + \frac{abm}{n} = \text{integer} \geq 0$$

$$h_{a+b+c}^{sc} - h_{a+b}^{sc} - h_{b+c}^{sc} - h_{a+c}^{sc} + h_a^{sc} + h_b^{sc} + h_c^{sc} = \text{even integer} \geq 0$$

- From the generalized Jacobi identity, we get a set a non-linear equations for $(h_a^{sc}, C_{ab}, \ldots)$
- Those condition may be sufficient and necessary $\rightarrow$ classification of Z_n-vertex algebra = classification of symmetric polynomials.
- Using Z_n-vertex algebra, we can calculate quasiparticle **scaling dimension**, quasiparticle **statistics**, **central charge (edge spectrum)**, ...

Topological order beyond FQH states

- A quantum state, a vector in $\mathcal{H}_{tot} = \otimes_i \mathcal{H}_i$:
  $|\Psi\rangle = \sum \Psi(m_1, ..., m_N)|m_1\rangle \otimes ... \otimes |m_N\rangle, \quad |m_i\rangle \in \mathcal{H}_i$
- A local Hamiltonian $H = \sum_{\mathbf{p}} O_{\mathbf{p}}$ and $O_{\mathbf{p}}$ are local hermitian operators acting within a few $\mathcal{H}_i$'s.
- A gapped Hamiltonian has following spectrum:



- Example: $H = \sum \sigma_i$, $\sigma_i = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ acts in $\mathcal{H}_i = \{|0\rangle, |1\rangle\}$.
  One ground state $|1\rangle \otimes ... \otimes |1\rangle$.
  It is a product state.

Two gapped states, $|\Psi(0)\rangle$ and $|\Psi(1)\rangle$, are in the same phase iff their Hamiltonian can be deformed into each other without closing the gap.

Quantum phases of matter and local unitary trans.

- Two gapped states, $|\Psi(0)\rangle$ and $|\Psi(1)\rangle$, are in the same phase iff they are related through a local unitary (LU) evolution

$$|\Psi(1)\rangle = P\Big(e^{-\mathrm{i}\int_0^1 dg\ \tilde{H}(g)}\Big)|\Psi(0)\rangle$$

where $\tilde{H}(g) = \sum_{\mathbf{i}} O_{\mathbf{i}}(g)$ and $O_{\mathbf{i}}(g)$ are local hermitian operators.

Hastings, Wen 05; Bravyi, Hastings, Michalakis 10

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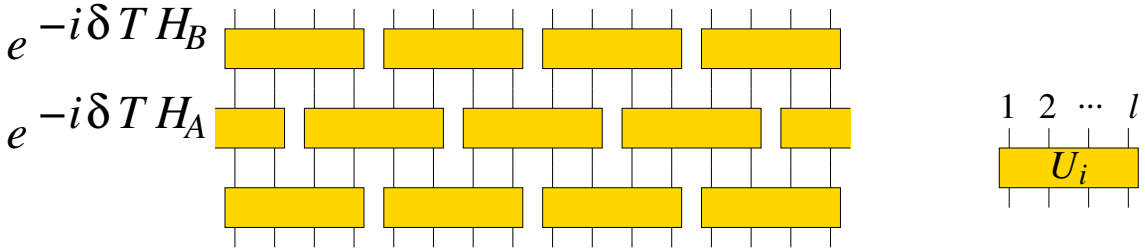
Hastings, Wen 05; Bravyi, Hastings, Michalakis 10

- LU evolution $P\left(e^{-\mathrm{i}\,T\int_0^1 dg\ \tilde{H}(g)}\right)$ = a quantum circuit = LU transformation



Topological order is a pattern of long range entanglement

- The local unitary transformations define an equivalence relation
  A universality class of a quantum phase is an equivalent class of the LU transformations

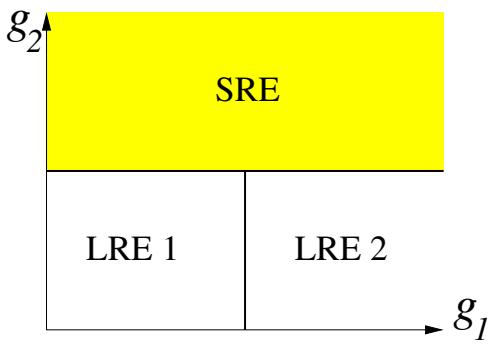
Topological order is a pattern of long range entanglement

- The local unitary transformations define an equivalence relation **A universality class of a quantum phase is an equivalent class of the LU transformations**

Two kinds of equivalent classes (phases) with no symmetries :

- The trivial class: states equivalent to product state under LU transformations. They belong to the same class (phase) $\rightarrow$ short-range entanglement and trivial topological order.
- Non-trivial classes: states not equivalent to direct-product states. Those states form many different equivalent classes (phases) $\rightarrow$ many patterns of long-range entanglements and many different topological orders.

Examples: FQH states



Labeling topological orders

Topological order = pattern of long range entanglement = equivalent class of LU transformations

How to label those equivalent classes?

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Labeling topological orders

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How to label those equivalent classes?

- We can use the wave function Φ itself to label the topological orders.
  But this is a many-many to one labeling scheme.

  Using the wave function renormalization, the wave function flows to simpler one within the same equivalent class.
- Use the fixed-point wave function: $\Phi \to \Phi_{\text{fix}}$ to label topological order.
  Hopefully Φ_{fix} can give us a one-to-one labeling of topological order, and a classification of topological order.

Local unitary trans. and wave function renormalization

- **The concept of support space**

  The total wave function $\Phi(i, j, ..., a, b, ...)$ can be viewed as a wave function on a region A: $\Phi_{a,b,...}(i, j, ...)$, where $i, j, ...$ label states in the region A and $a, b, ...$ label states out side of the region A.

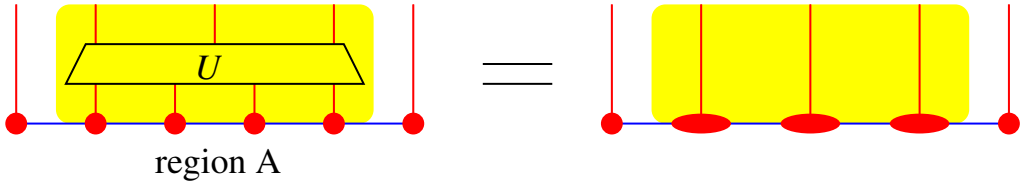
  $\Phi_{a,b,...}(i, j, ...)$ for all different $a, b, ...$ span the support space $\mathcal{H}_A^{sp}$ of the region A.

  The dimension of the support space is less then the total Hilbert space $\mathcal{H}_A$ in A: $\mathcal{H}_A^{sp} \subset \mathcal{H}_A = \otimes_{i \in A} \mathcal{H}_i$

- **The concept of generalized unitary transformation**

  A generalized unitary transformation $U: \mathcal{H}_A \to \mathcal{H}_A^{sp}$ shrinks the degrees of freedom in A *without loosing any quantum information*. It generates a wave function renormalization:

Vidal 07; Jordan, Orus, Vidal, Verstraete, Cirac 08; Jiang, Weng, Xiang 09; Gu, Levin, Wen 09

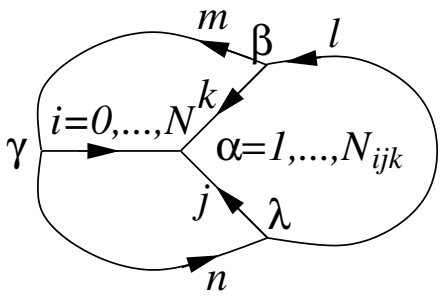


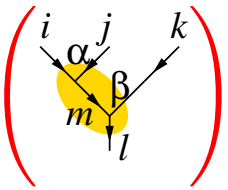
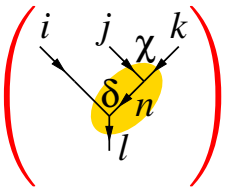
The first kind of wave function renormalization: F-move

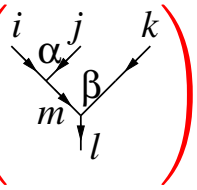
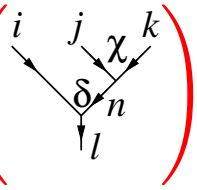
Transformation properties of fixed-point wave function:

- The fixed-point wave functions are related by a fixed-point LU trans.:

$$\Phi(\text{graph-2}) = U\Phi(\text{graph-1}),$$



- The two wave functions $\Phi =$  and $\Phi =$  on different graphs are related by a LU trans. Levin, Wen, 04

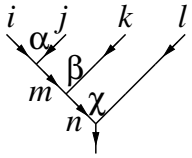
F-move: Φ  $= \sum_{n=0}^{N} \sum_{\chi=1}^{N_{kjn^*}} \sum_{\delta=1}^{N_{nil^*}} F^{ijm,\alpha\beta}_{kln,\chi\delta}\, \Phi$ 

Two important data:

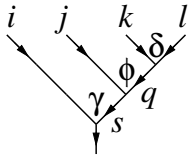
The unitary matrix $F^{ij}_{kl} \to (F^{ij}_{kl})^{m,\alpha\beta}_{n,\chi\delta}$

The dimension of support space N_{ijk}

The pentagon identity



can be trans. to



through two different paths:

$$\Phi\left(\begin{array}{c}\text{diagram}\end{array}\right) = \sum_{q,\delta,\epsilon} F^{mkn,\beta\chi}_{lpq,\delta\epsilon}\,\Phi\left(\begin{array}{c}\text{diagram}\end{array}\right) = \sum_{q,\delta,\epsilon;s,\phi,\gamma} F^{mkn,\beta\chi}_{lpq,\delta\epsilon}\,F^{ijm,\alpha\epsilon}_{qps,\phi\gamma}\,\Phi\left(\begin{array}{c}\text{diagram}\end{array}\right)$$

$$\Phi\left(\begin{array}{c}\text{diagram}\end{array}\right) = \sum_{t,\eta,\varphi} F^{ijm,\alpha\beta}_{knt,\eta\varphi}\,\Phi\left(\begin{array}{c}\text{diagram}\end{array}\right) = \sum_{t,\eta,\varphi;s,\kappa,\gamma} F^{ijm,\alpha\beta}_{knt,\eta\varphi}\,F^{itn,\varphi\chi}_{lps,\kappa\gamma}\,\Phi\left(\begin{array}{c}\text{diagram}\end{array}\right)$$

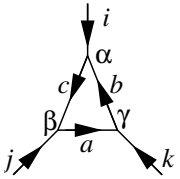
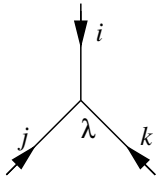
$$= \sum_{t,\eta,\kappa;\varphi;s,\kappa,\gamma;q,\delta,\phi} F^{ijm,\alpha\beta}_{knt,\eta\varphi}\,F^{itn,\varphi\chi}_{lps,\kappa\gamma}\,F^{jkt,\eta\kappa}_{lsq,\delta\phi}\,\Phi\left(\begin{array}{c}\text{diagram}\end{array}\right).$$

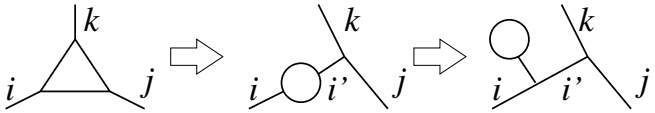
The two paths should lead to the same LU trans.:

$$\sum_{t,\eta,\varphi,\kappa} F^{ijm,\alpha\beta}_{knt,\eta\varphi}\,F^{itn,\varphi\chi}_{lps,\kappa\gamma}\,F^{jkt,\eta\kappa}_{lsq,\delta\phi} = \sum_{\epsilon} F^{mkn,\beta\chi}_{lpq,\delta\epsilon}\,F^{ijm,\alpha\epsilon}_{qps,\phi\gamma}$$

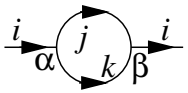
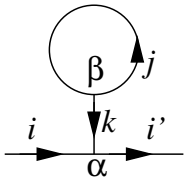
Such a set of non-linear algebraic equations is the famous pentagon identity.

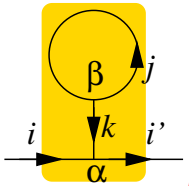
The second kind of wave function renormalization: P-move

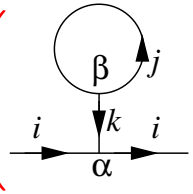
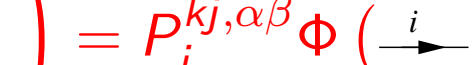
- First way:  to  which reduce the degrees of freedom.

  But notice that, through the F-move, 

- Second way:

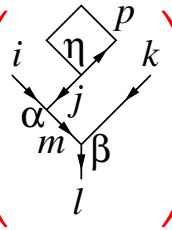
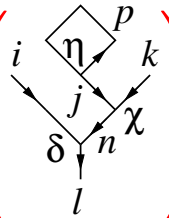
   to Levin, Wen, 04 or  to Koenig, Reichardt, Vidal, 09

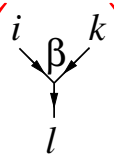
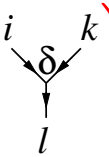
  The support dimension of $\Phi\Big($  $\Big)$: $D_{ii'*} = \delta_{ii'}$

  $\Phi\Big($  $\Big) = P_i^{kj,\alpha\beta}\Phi\Big($  $\Big)$, $\displaystyle\sum_{\alpha=1}^{N_{kii^*}}\sum_{\beta=1}^{N_{j^*jk^*}} P_i^{kj,\alpha\beta}(P_i^{kj,\alpha\beta})^* = 1$

Consistent conditions between $F^{ijm,\alpha\beta}_{kln,\chi\delta}$ and $P^{kj,\alpha\beta}_{i}$

From

$\Phi\Big($  $\Big) = \sum_{n=0}^{N}\sum_{\chi,\delta} F^{ijm,\alpha\beta}_{kln,\chi\delta}\,\Phi\Big($  $\Big)$

$\rightarrow \quad P^{jp,\alpha\eta}_{i}\,\delta_{im}\,\Phi\Big($  $\Big) = \sum_{n,\chi,\delta} F^{ijm,\alpha\beta}_{kln,\chi\delta}\,P^{jp,\chi\eta}_{k^*}\,\delta_{kn}\,\Phi\Big($  $\Big)$

we find more non-linear equation

$$P^{jp,\alpha\eta}_{i}\,\delta_{im}\,\delta_{\beta\delta} = \sum_{\chi=1}^{N_{kjk^*}} F^{ijm,\alpha\beta}_{klk,\chi\delta}\,P^{jp,\chi\eta}_{k^*}$$

for all k, i, l satisfying $N_{kil^*} > 0$

A classification of topological orders by tensor category

The data $N_{ijk}, F^{ijm,\alpha\beta}_{kln,\chi\delta}, P^{kj,\alpha\beta}_{i}$ classify the fixed-point LU transformation and topological orders: Levin-Wen 05; Chen-Gu-Wen 10

$$\sum_m N_{jim^*} N_{kml^*} = \sum_n N_{kjn^*} N_{l^* ni},$$

$$(F^{ijm,\alpha\beta}_{kln,\chi\delta})^* = F^{jkn,\chi\delta}_{l^* i^* m^*,\beta\alpha},$$

☞ $$\sum_{n,\chi,\delta} F^{ijm',\alpha'\beta'}_{kln,\chi\delta} (F^{ijm,\alpha\beta}_{kln,\chi\delta})^* = \delta_{m\alpha\beta,m'\alpha'\beta'},$$

☞ $$\sum_{t,\eta,\varphi,\kappa} F^{ijm,\alpha\beta}_{knt,\eta\varphi} F^{itn,\varphi\chi}_{lps,\kappa\gamma} F^{jkt,\eta\kappa}_{lsq,\delta\phi} = \sum_{\epsilon} F^{mkn,\beta\chi}_{lpq,\delta\epsilon} F^{ijm,\alpha\epsilon}_{qps,\phi\gamma},$$

☞ $$\sum_{\alpha=1}^{N_{kii^*}} \sum_{\beta=1}^{N_{j^* jk^*}} P^{kj,\alpha\beta}_{i} (P^{kj,\alpha\beta}_{i})^* = 1,$$

$$P^{kj,\alpha\beta}_{i} = \sum_{m,\lambda,\gamma,l,\nu,\mu} F^{jj^* k,\beta\alpha}_{i^* i^* m^*,\lambda\gamma} F^{i^* mj^*,\lambda\gamma}_{m^* i^* l,\nu\mu} P^{lm,\mu\nu}_{i^*},$$

☞ $$P^{jp,\alpha\eta}_{i} \delta_{im} \delta_{\beta\delta} = \sum_{\chi} F^{ijm,\alpha\beta}_{klk,\chi\delta} P^{jp,\chi\eta}_{k^*} \quad \text{for all } k, i, l \text{ with } N_{kil^*} > 0.$$

Spherical fusion category and its corresponding MTC

- The data $N_{ijk}, F^{ijm,\alpha\beta}_{kln,\chi\delta}, P^{kj,\alpha\beta}_{i}$ describe a spherical fusion category
- The fixed point state has quasiparticle excitations described by a MTC – The Drinfeld center of the spherical fusion category.

- In FQH state: pattern of zero $\rightarrow$ MTC that describes the quasiparticles
- In fixed-point state: spherical fusion category $\rightarrow$ MTC that describes the quasiparticles

Classify 2D fermionic topological order

The non-chiral 2D **fermionic** topological orders are (partially?) classified by another set of tensors $N_{ijk}, N^f_{ijk}, F^{ijm,\alpha\beta,\pm}_{kln,\gamma\lambda,\pm}, O^{jk,\alpha\beta}_{i,\pm}, A^i$ that satisfy Gu-Wang-Wen 10

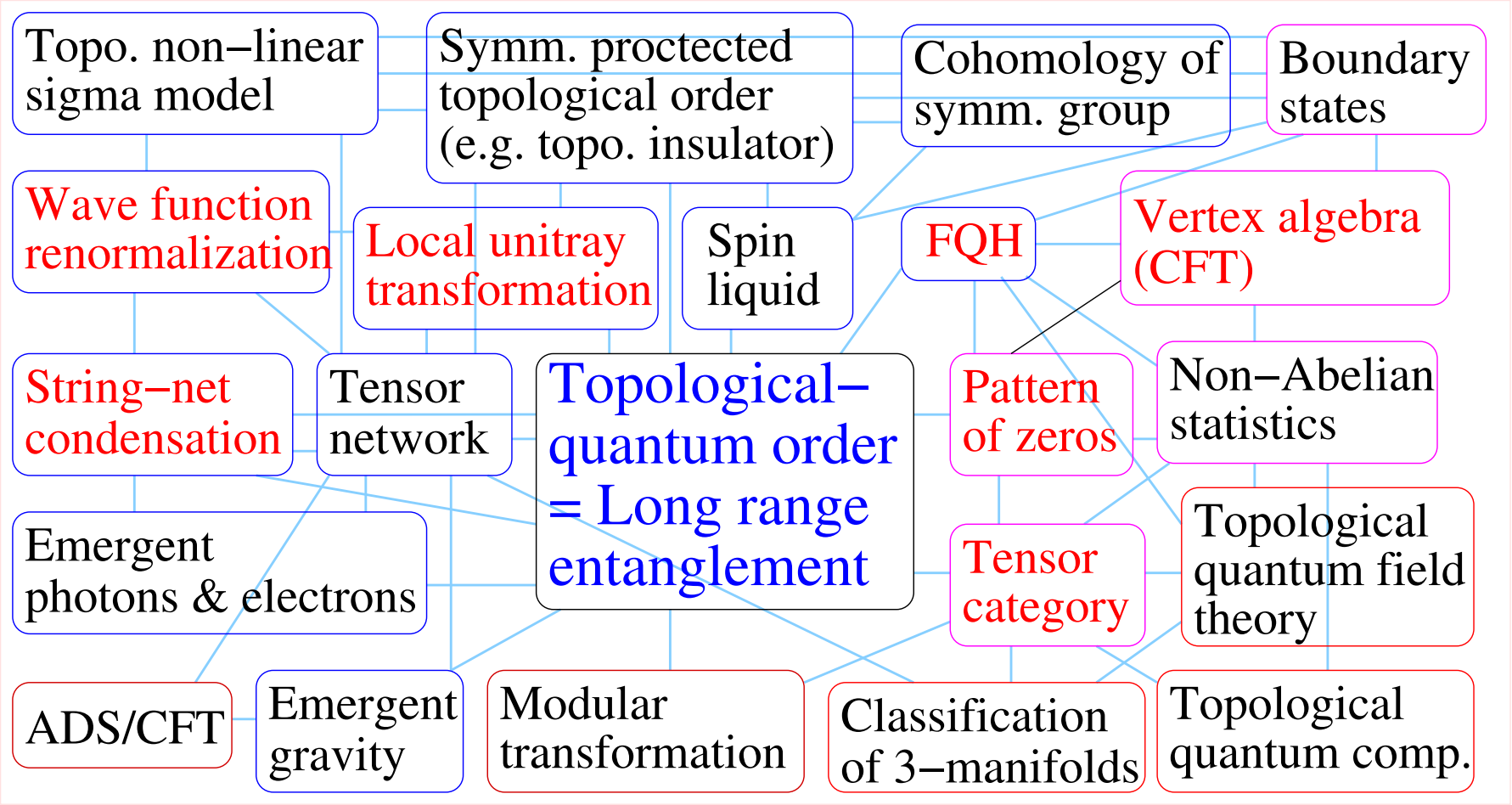
☞ $$\sum_{m=0}^{N} N_{jim^*} N_{kml^*} = \sum_{n=0}^{N} N_{kjn^*} N_{l^* ni},$$

☞ $$\sum_{m=0}^{N} (N^b_{jim^*} N^f_{kml^*} + N^f_{jim^*} N^b_{kml^*}) = \sum_{n=0}^{N} (N^b_{kjn^*} N^f_{l^* ni} + N^f_{kjn^*} N^b_{l^* ni}),$$

☞ $$\sum_{t} \sum_{\eta=1}^{N_{kjt^*}} \sum_{\varphi=1}^{N_{tin^*}} \sum_{\kappa=1}^{N_{lts^*}} F^{ijm,\alpha\beta,+}_{knt,\eta\varphi,-} F^{itn,\varphi\chi,+}_{lps,\kappa\gamma,-} F^{jkt,\eta\kappa,+}_{lsq,\delta\phi,-}$$

$$= (-)^{s_{jim^*}(\alpha) s_{lkq^*}(\delta)} \sum_{\epsilon=1}^{N_{qmp^*}} F^{mkn,\beta\chi,+}_{lpq,\delta\epsilon,-} F^{ijm,\alpha\epsilon,+}_{qps,\phi\gamma,-}$$

.....

- This is a super spherical fusion category?
  $\rightarrow$ a super MTC for its quasiparticle?

Highly entangled state: A rich world of topo. order and entanglement